

Approach 0: Full Context Dump

Provide every past thing as context.
Works initially. Remembers everything.



Bottleneck

Context grows huge → chat hangs.
Retrieval from long texts becomes difficult →
important things get lost or de-prioritized.



Approach 1: Sliding Window

Take only the past 'N' messages as context
and keep this sliding window.
Uses recent & updated info; avoids long context issues.



Bottleneck

Discards old conversation altogether.
Important things in old conversations are lost.
Leads to results not aligned with initial plans.



Approach 2: Naive Retrieval & Semantic Similarity

Extract and store key "nuggets" (decisions, updates).
Embed incoming requests & stored nuggets.
Retrieve relevant info via semantic similarity search.



Bottleneck

Naive retrieval: everything extracted gets stored (no write-side judgment).
Similarity-only ranking: no extra importance for recency
or how often something's been useful.
No mechanism to handle conflicts or ambiguity.



→ Derived Capabilities

See first_principles.md